Multiple Linear Regression Analysis

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots \beta_k x_k + u$$

Estimation

Some Terminology

- Like the simple regression model, in multiple linear regression models, we typically call

y	$x_1, x_2, \ldots, x_k$
Dependent Variable	Independent Variable
Left-Hand Side Variable	Right-Hand Side Variable
Explained Variable	Explanatory Variable
Regressand	Regressor
	Covariate, Control Variables

Motivation for multiple regression

- Again, we usually make ***Zero Conditional Mean Hypothesis,*** the same with the single regressor model: $E(u|x) = 0$, which implies here:

  $E(y|x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots \beta_k x_k$
- $E(u|x) = 0$ implies that the regression disturbance u contains no systematic information of x that can be used to predict the mean of y.

Motivation for multiple regression

For simple model: $y = \beta_0 + \beta_1 educ + u,$

It's unlikely that the condition $E(u \mid educ) = 0$ holds.

Q: Why?

A: Many important variables are omitted in u.

Thus a multi model: $y = \beta_0 + \beta_1 educ + \beta_2 experience + u,$

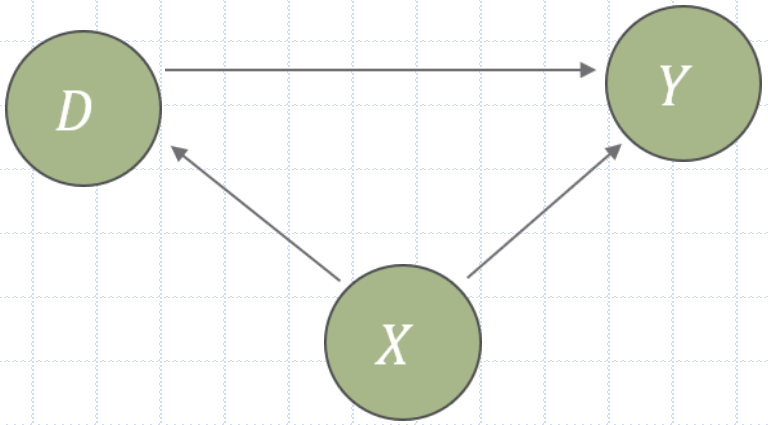
where $E(u \mid educ, experience) = 0,$

may be more reasonable.

Motivation for multiple regression

- Why would we want to include multiple variables in a regression?

Y: **Earnings**
D: **Education**
X: **Experience**
Affects both *D* and *Y*!



Motivation for multiple regression

- We know that $D \rightarrow Y$, but there's another open channel: $D \leftarrow X \rightarrow Y$.
- To close this channel, we want to "control for" experience or other related factors.

- Now, what's the economic meaning of β_1.

Obtaining the OLS Estimates

- OLS (minimizing SSR):

$$\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik})^2$$

- This minimization problem can be solved using multivariable calculus.
- The solution is denoted as $(\hat{\beta}_0, \hat{\beta}_1, \cdots, \hat{\beta}_k)'$ as OLS estimate of $(\beta_0, \beta_1, \cdots, \beta_k)'$.

Obtaining the OLS Estimates

- This lead to k+1 linear equation in k+1 unknows $\hat{\beta}_0, \hat{\beta}_1, \cdots, \hat{\beta}_k$:

$$\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}) = 0$$

$$\sum_{i=1}^{n} x_{i1}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}) = 0$$

$$\sum_{i=1}^{n} x_{i2}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}) = 0$$

$$\vdots$$

$$\sum_{i=1}^{n} x_{ik}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}) = 0.$$

Obtaining the OLS Estimates

- As with the simple regression model, these are often called the OLS first order conditions: the sample counterparts of these population moments, although we have omitted the division by the sample size n.
- For even moderately sized n and k, solving the equations by hand calculations is tedious, but econometrics software can solve these equations with large n and k very quickly.

Interpretation of the OLS Estimates

- We begin with the case of two independent variables:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$$

- The estimates $\hat{\beta}_1$ and $\hat{\beta}_2$ have **partial effect**, or **ceteris paribus**, interpretation:

$$\Delta\hat{y} = \hat{\beta}_1 \Delta x_1 + \hat{\beta}_2 \Delta x_2$$

- If, for example, x_2 is held fixed so that $\Delta x_2 = 0$:

$$\Delta\hat{y} = \hat{\beta}_1 \Delta x_1$$

Interpretation of the OLS Estimates

- The key point is that, by including x_2 in our model, we obtain a coefficient on x_1 with a ceteris paribus interpretation. This is why multiple regression analysis is so useful.
- Similarly, if x_1 is held fixed so that $\Delta x_1 = 0$:

$$\Delta \hat{y} = \hat{\beta}_2 \Delta x_2$$

Interpretation of the OLS Estimates

- The case with more than two independent variables is similar.

$$\Delta\hat{y} = \hat{\beta}_1\Delta x_1 + \cdots + \hat{\beta}_k\Delta x_k$$

- The coefficient on x_1 measures the change in $\hat{y}$ due to a one-unit increase in x_1, holding all other independent variables fixed:

$$\Delta\hat{y} = \hat{\beta}_1\Delta x_1$$

- Thus, we also say that we have **controlled for** the variables $x_2, x_3, \cdots x_k$ when estimating the effect of x_1 on y.

A "Partialling Out" Interpretation

- When applying OLS, for certain derivations, we do need explicit formulas for the $\hat{\beta}_j$. These formulas also shed further light on the workings of OLS.
- Consider again the case with k=2 independent variables:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$$

A "Partialling Out" Interpretation

- For concreteness, we focus on $\hat{\beta}_1$. One way to express $\hat{\beta}_1$ is:

$$\hat{\beta}_1 = \left(\sum_{i=1}^{n} \hat{r}_{i1} y_i\right) \Big/ \left(\sum_{i=1}^{n} \hat{r}_{i1}^2\right),$$

  - where the $\hat{r}_{i1}$ are the OLS residuals from a simple regression of x_1 on x_2.
- This expression gives another demonstration of $\hat{\beta}_1$'s partial interpretation !

"Partialling Out" continued

- Specifically, $\hat{r}_{i1}$ is the part of x_1 that is uncorrelated with x_2 , so we're estimating the effect of x_1 on y after x_2 has been "partialled out".
- This implies that regressing y on x_1 and x_2 gives same effect of x_1 as regressing y on residuals from a regression of x_1 on x_2.

Simple vs Multiple Reg Estimate

Compare the simple regression $\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1$

with the multiple regression $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$

thus $\tilde{\beta}_1 = \hat{\beta}_1 + \hat{\beta}_2 \tilde{\delta}_1$, where $\tilde{\delta}_1$ is the slope coefficient(x_2 on x_1)

Generally, $\tilde{\beta}_1 \neq \hat{\beta}_1$ unless:

(1) $\hat{\beta}_2 = 0$ i.e. no partial effect of x_2.

(2) $\tilde{\delta}_1 = 0$ i.e. x_1 and x_2 are uncorrelated in the sample.

Goodness-of-Fit

- How do we think about how well our sample regression line fits our sample data?
- Can compute the fraction of the total sum of squares (SST) that is explained by the model, call this the R-squared of regression

$$R^2 = SSE/SST = 1 - SSR/SST$$

More about *R*-squared

- In multiple regression, R^2 can never decrease when another independent variable is added to a regression, and usually will increase.
- R^2 increases with the addition of independent variables, even if the newly added independent variables make no substantive contribution to the model, which may lead to overfitting.

Adjusted R^2

- The adjusted R^2 or $\bar{R}^2$ takes into account the number of variables and sample size (give penalty on adding variables), thereby avoiding the problem of overfitting:

$$\bar{R}^2 = 1 - \frac{n-1}{n-k-1} \times \frac{SSR}{SST}$$

- The difference with R^2 is that the ratio of SSR to SST is multiplied by the factor $(n\text{-}1)/(n\text{-}k\text{-}1)$.

Adjusted R^2

- When adding a regressor, there are two useful things to know about the $\bar{R}^2$:
  - ✓ Since $(n-1)/(n-k-1)$ always greater than 1, so $\bar{R}^2$ is always less than R^2.
  - ✓ Whether the $\bar{R}^2$ increase or decrease depends on two opposite effects: the SSR falls, will increase the $\bar{R}^2$; the factor $(n-1)/(n-k-1)$ increases.

Adjusted R^2

- Nevertheless, heavy reliance on the $\bar{R}^2$ or R^2 can be a trap.
- The decision about whether to include a variable in a multiple regression should be based on whether including that variable allows you better to estimate the causal effect of interest.
- Recall that R^2 says nothing about $E(u|x) = 0$!

Unbiasedness of OLS estimator

A1. Population model is ***linear*** in parameters: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k + u$

A2. We can use a ***random sample*** of size n, $\{(x_{i1}, x_{i2}, \ldots, x_{ik}, y_i)$: $i=1, 2, \ldots, n\}$, from the population model, so that the population model is $y_i = \beta_0 + \beta_1 x_{i1} + \ldots + \beta_k x_{ik} + u_i$

A3. $E(u|x_1, x_2, \ldots x_k) = 0$, implying that all of the explanatory variables are ***conditional exogenous.***

Unbiasedness of OLS estimator

A4. None of the x's is constant, and there are ***no exact linear relationships*** among them.

Under A1-A4, OLS estimator is unbiased, i.e.,

$$\mathrm{E}(\hat{\beta}) = \beta.$$

Too Many Variables and bias

- What happens if we include irrelevant variables?

  ANS: There is no effect on the biasedness of our parameter estimate, and OLS remains unbiased

Too Few Variables and bias

- What if we exclude a variable from our specification that does belong?

  ANS: OLS will usually be biased, i.e., $E(\hat{\beta}) \neq \beta$.

Omitted Variable Bias

- Suppose the true model is given as:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$$

- But we mistakenly estimate $y_i = \alpha_0 + \alpha_1 x_{i1} + e_i$, then

$$\alpha_1 = \frac{\sum (x_{i1} - \bar{x}_1)\, y_i}{\sum (x_{i1} - \bar{x}_1)^2}$$

Omitted Variable Bias

- Take the y_i into numerator, we can get:

$$\alpha_1 = \beta_1 + \beta_2 \frac{\sum(x_{i1}-\bar{x}_1)x_{i2}}{\sum(x_{i1}-\bar{x}_1)^2} + \frac{\sum(x_{i1}-\bar{x}_1)\varepsilon_i}{\sum(x_{i1}-\bar{x}_1)^2}$$

Since $E(\varepsilon_i|x) = 0$, taking expectations we have:

$$E(\alpha_1|x) = \beta_1 + \beta_2 \frac{\sum(x_{i1} - \bar{x}_1)\, x_{i2}}{\sum(x_{i1} - \bar{x}_1)^2}$$

- The last term $(E(\alpha_1|x) - \beta_1)$ is also called **omitted variable bias**.

Omitted Variable Bias Summary

- Two cases where bias is equal to zero:
  - ✓ $\beta_2 = 0$, that is x_2 does not really belong in the model
  - ✓ x_1 and x_2 are uncorrelated in the sample
- Otherwise, the direction of bias will be written as:

	$Corr(x_1, x_2) > 0$	$Corr(x_1, x_2) < 0$
$\beta_2 > 0$	positive bias	negative bias
$\beta_2 < 0$	negative bias	positive bias

Ex: hourly wage equation

Suppose

$$\ln(wage) = \beta_0 + \beta_1 edu + \beta_2 abil + u,$$

we don't have data on ability, so we estimate

$$\ln(\hat{w}age) = 0.584 + 0.083 edu.$$

Q: Compared to the true return to education,

would 0.083 be too large or too small from a statistical view?

The More General Case

Suppose

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$$

satisfies assumption A1-4, but we omit x_3 and estimate

$$\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1 + \tilde{\beta}_2 x_2$$

Q: Suppose x_2 and x_3 are uncorrelated, but x_1 and x_3 are not, will $\tilde{\beta}_1$ and $\tilde{\beta}_2$ be unbiased?

A: Although x_2 is uncorrelated with x_3, $\tilde{\beta}_2$ is also biased. Unless x_2 and x_1 are also uncorrelated.

Variance of the OLS Estimators

- Now we know that the sampling distribution of our estimate is centered around the true parameter
- Want to know about the spread in its sampling distribution
- Much easier to think about this variance under an additional assumption, so

A5. Assume $\text{Var}(u|x_1, x_2, \ldots, x_k) = \sigma^2$ (Conditional Homoskedasticity)

Variance of OLS (cont)

- Let $\boldsymbol{x}$ stand for $(x_1, x_2, \ldots x_k)$
- Assuming that $\text{Var}(u|\boldsymbol{x}) = \sigma^2$ also implies that $\text{Var}(y|\boldsymbol{x}) = \sigma^2$
- The 4 assumptions for unbiasedness, plus this homoskedasticity assumption are known as the ***Gauss-Markov*** assumptions

Variance of OLS (cont)

Given the Gauss-Markov Assumptions,
the variance of each estimated parameter is

$$Var\left(\hat{\beta}_j \mid x\right) = \frac{\sigma^2}{SST_j\left(1 - R_j^2\right)},$$

where $SST_j = \sum\left(x_{ij} - \bar{x}_j\right)^2$

and R_j^2 is the R^2 from regressing x_j on all other x's.

Components of OLS Variances

- The error variance: a larger σ^2 (noise) implies a larger variance for the OLS estimators
- The total sample variation: a larger SST_j implies a smaller variance for the estimators
  - ✓ Variation in x_j is a good thing!
  - ✓ Increase sample size is also good!
- Linear relationships among the independent variables: a larger R_j^2 implies a larger variance for the estimators

Components of OLS Variances

- When $R_j^2 = 1$, x_j is the perfect linear combination of some of the other independent variables.

$$\text{Var}(\hat{\beta}_j \mid x) \rightarrow \infty \text{ as } R_j^2 \rightarrow 1$$

- High (but not perfect) correlation is called *Multicollinearity*.

Exp: Multicollinearity

- For the following regression model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$$

- Suppose that x_2 and x_3 are highly correlated, but they are not highly correlated with x_1. Now if we are *only* interested in var($\hat{\beta}_1$|x), it doesn't matter.

Misspecified Models

Suppose the true model

$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2,$

if we misspecified it as

$\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1,$

then $Var\left(\tilde{\beta}_1\right) = \dfrac{\sigma^2}{SST_1}, \quad Var\left(\hat{\beta}_1\right) = \dfrac{\sigma^2}{SST_1\left(1 - R_1^2\right)},$

case 1: if x_1 and x_2 are correlated, $Var\left(\tilde{\beta}_1\right) < Var\left(\hat{\beta}_1\right),$

case 2: if x_1 and x_2 are uncorrelated, then they're the same.

Estimating the Error Variance

- Again, we don't know what the error variance, σ^2, is, because we don't observe the errors, u_i
- What we observe are the residuals, $\hat{u}_i$
- We can use the residuals to form an estimate of the error variance, as we have done in the simple regression model.

Error Variance Estimator (cont)

$$\hat{\sigma}^2 = \left(\sum \hat{u}_i^2\right)/(n-k-1) \equiv SSR/df$$

$$\text{thus, } se\left(\hat{\beta}_j\right) = \hat{\sigma}/\left[SST_j\left(1-R_j^2\right)\right]^{1/2}$$

- Degrees of freedom:

  $df = n - (k + 1)$, or $df = n - k - 1$

  i.e. the (number of observations) – (number of estimated parameters)

The Gauss-Markov Theorem

- Given our 5 Gauss-Markov Assumptions it can be shown that OLS is "BLUE"
  - Best
  - Linear
  - Unbiased
  - Estimator
- Thus, if the assumptions hold, use OLS

Summary

- Multiple regression model:
  - ceteris paribus
  - omitted variable bias
- OLS: unbiasedness and variance
- OLS: efficiency (Gauss-Markov theorem)

- Next time: Inference and Testing Hypotheses